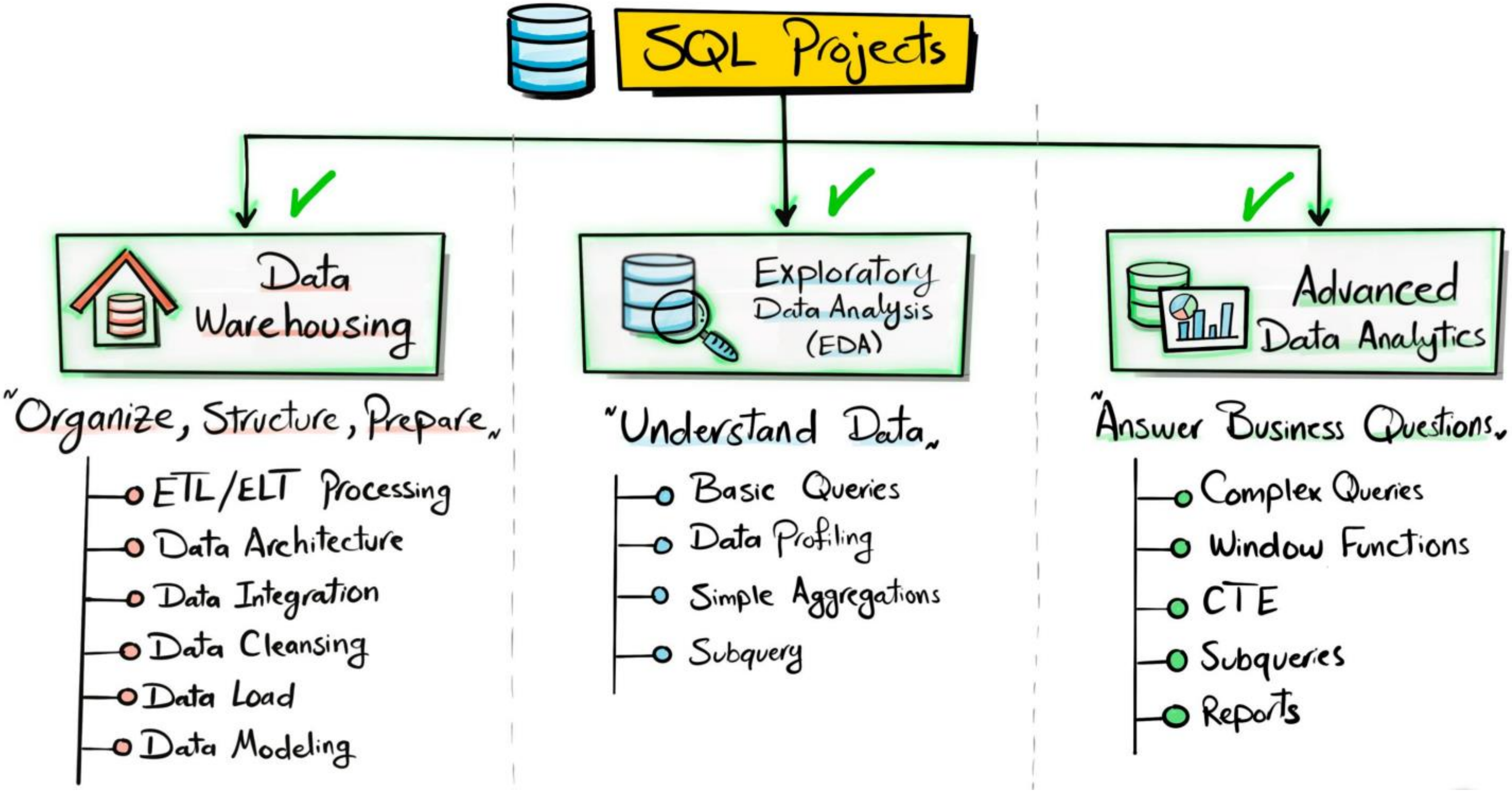
SQL DATA Analytics

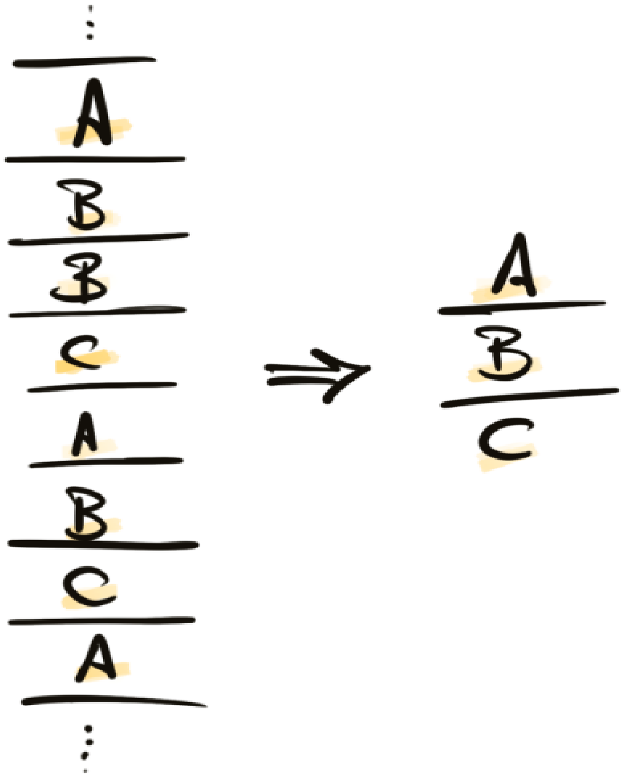
Project


SQL Projects
Data Warehousing
Exploratory Data Analysis (EDA)
Advanced Data Analytics
"Organize, Structure, Prepare"
ETL/ELT Processing
Data Architecture
Data Integration
Data Cleansing
Data Load
Data Modeling
"Understand Data"
Basic Queries
Data Profiling
Simple Aggregations
Subquery
"Answer Business Questions"
Complex Queries
Window Functions
CTE
Subqueries
Reports

Dimensions Exploration

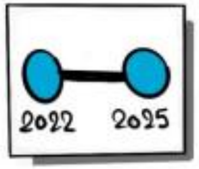
DISTINCT [Dimension]

DISTINCT Country
DISTINCT Category
DISTINCT Product



Activity - Dimensions Exploration
Q1. Retrieve a list of unique countries from which customers originate.
Q2. Retrieve a list of unique categories, subcategories, and products

Purpose:
To explore the structure of dimension tables.
SQL Functions Used: DISTINCT & ORDER BY



Date Exploration

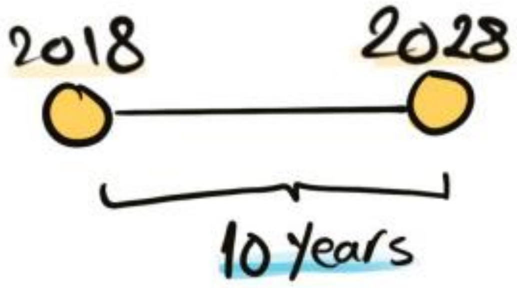
MIN/MAX [Date Dimension]

MIN Order_date

MAX Create_date

MIN Birthdate

2019
2020
2018
2018
2022
2023
2023
2028
2022



DATEDIFF

Activity - Date Range Exploration

Q1. Determine the first and last order date and the total duration in months

Q2. Find the youngest and oldest customer based on birthdate

Purpose:

To determine the temporal boundaries of key data points.

To understand the range of historical data.

SQL Functions Used:- MIN(), MAX(), DATEDIFF()

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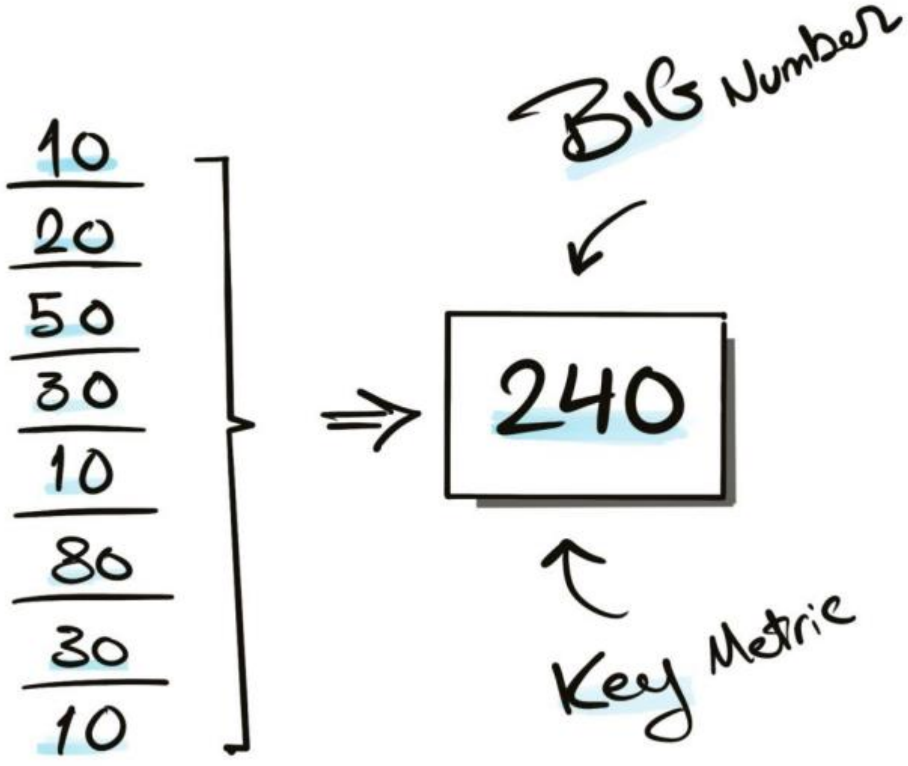
Measures Exploration

∑ [Measure]

SUM (Sales)
AVG (Price)
SUM (Quantity)



Activity - Measures Exploration (Key Metrics)
Q1. Find the Total Sales,
Q2. Find how many items are sold
Q3. Find the average selling price
Q4. Find the Total number of Orders
Q5. Find the total number of products
Q6. Find the total number of customers
Q7. Find the total number of customers that has placed an order
Q8. Generate a Report that shows all key metrics of the business
Purpose:
To calculate aggregated metrics (e.g., totals, averages) for quick insights.
To identify overall trends or spot anomalies.
SQL Functions Used: COUNT(), SUM(), AVG()

Magnitude

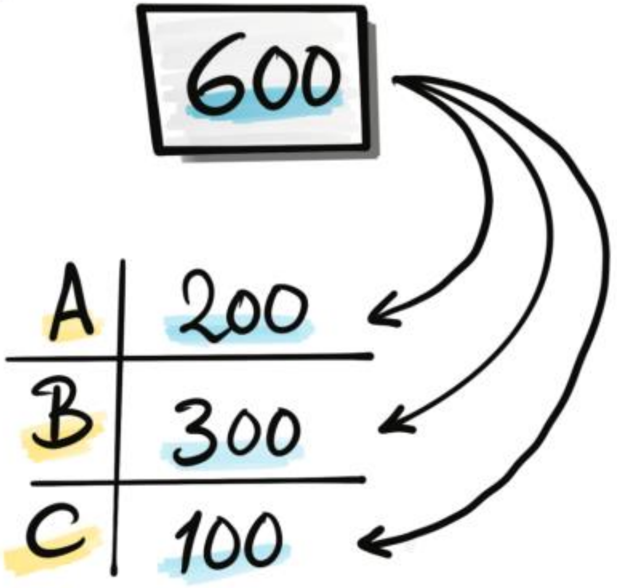
$\sum$ [Measure] By [Dimension]

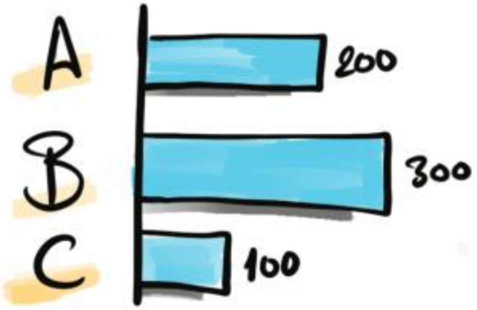
Total Sales By Country

Total Quantity By Category

Average Price By Product

Total Orders By Customer





Activity : Magnitude Analysis
Q1. Find total customers by countries.
Q2. Find total customers by gender
Q3. Find total products by category
Q4. What is the average costs in each category?
Q5. What is the total revenue generated for each category?
Q6. What is the total revenue generated by each customer?
Q7. What is the distribution of sold items across countries?

Purpose:
To quantify data and group results by specific dimensions.
For understanding data distribution across categories.
SQL Functions Used: SUM, COUNT, AVG, GROUP BY, ORDER BY

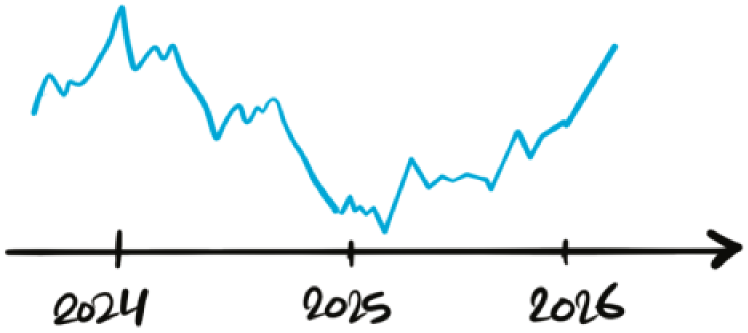
Change-Over-Time Trends

$\sum$ [Measure] By [Date Dimension]

Total Sales By Year

Average Cost By Month

2024	300
2025	100
2026	200



Activity - Analyze sales performance over time- Via Date Functions

Q1. Find the yearly & Monthly Sales with total customer count.
Q2. Find the yearly & Monthly Sales with total customer count. (Using DATETRUNC)
Q3. Find the yearly & Monthly Sales with total customer count. FORMAT()

Purpose:

- To track trends, growth, and changes in key metrics over time.
- For time-series analysis and identifying seasonality.
- To measure growth or decline over specific periods.

SQL Functions Used:

- Date Functions: DATEPART(), DATETRUNC(), FORMAT()
- Aggregate Functions: SUM(), COUNT(), AVG()

Cumulative Analysis

∑ [Cumulative Measure] By [Date Dimension]

WINDOW FUNCTIONS

Running Total Sales By Year

Moving Average of Sales By Month

Cumulative

2024	300	300
2025	100	400
2026	200	600

Activity - Cumulative Analysis

Q1. Calculate the total sales per month
Q2. Also, the running total of sales over time

Purpose:
- To calculate running totals or moving averages for key metrics.
- To track performance over time cumulatively.
- Useful for growth analysis or identifying long-term trends.

SQL Functions Used:
- Window Functions: SUM() OVER(), AVG() OVER()

Performance Analysis

WINDOW FUNCTIONS

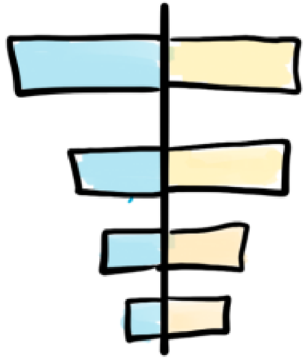
Current [Measure] – Target [Measure]

Current Sales – Average Sales

Current Year Sales – Previous Year Sales

Current Sales – lowest Sales

	Current	Target (AVG)	Performance
A	200	200	0
B	300	200	100
C	100	200	-100



Activity - Performance Analysis (Year-over-Year, Month-over-Month)

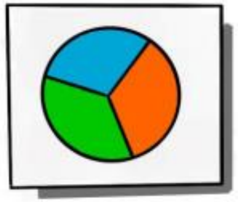
Analyze the yearly performance of products by comparing their sales to both the average sales performance of the product and the previous year's sales.
Like Table: Year, Product Name, Current Sales, Average Sales, Difference (Current Sales – Average Sales), Avg Change flag (Above avg./Below Avg.), Prv. Year sales, Difference (Current sales – Prv Sales), Yearly Change (Increase/decrease/No change)

Purpose:

- To measure the performance of products, customers, or regions over time.
- For benchmarking and identifying high-performing entities.
- To track yearly trends and growth.

SQL Functions Used:

- LAG(): Accesses data from previous rows.
- AVG() OVER(): Computes average values within partitions.
- CASE: Defines conditional logic for trend analysis.



Part-to-Whole

Proportional Analysis

Activity 1 - Part-to-Whole Analysis

Q1. Which categories contribute the most to overall sales? (%wise category sales)

Purpose:
- To compare performance or metrics across dimensions or time periods.
- To evaluate differences between categories.
- Useful for A/B testing or regional comparisons.

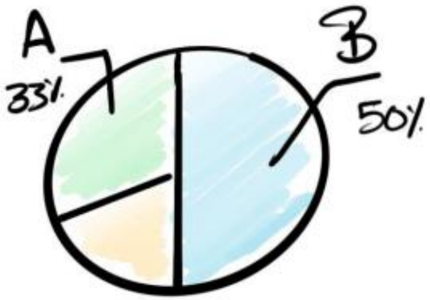
SQL Functions Used:
- SUM(), AVG(): Aggregates values for comparison.
- Window Functions: SUM() OVER() for total calculations.

([Measure] / Total [Measure]) * 100 By [Dimension]

(Sales / Total Sales) * 100 By Category

(Quantity / Total Quantity) * 100 By Country

A	200	33%
B	300	50%
C	100	17%

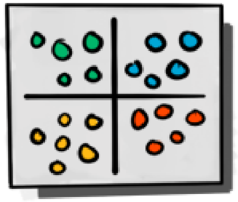


Activity 2 - Ranking Analysis

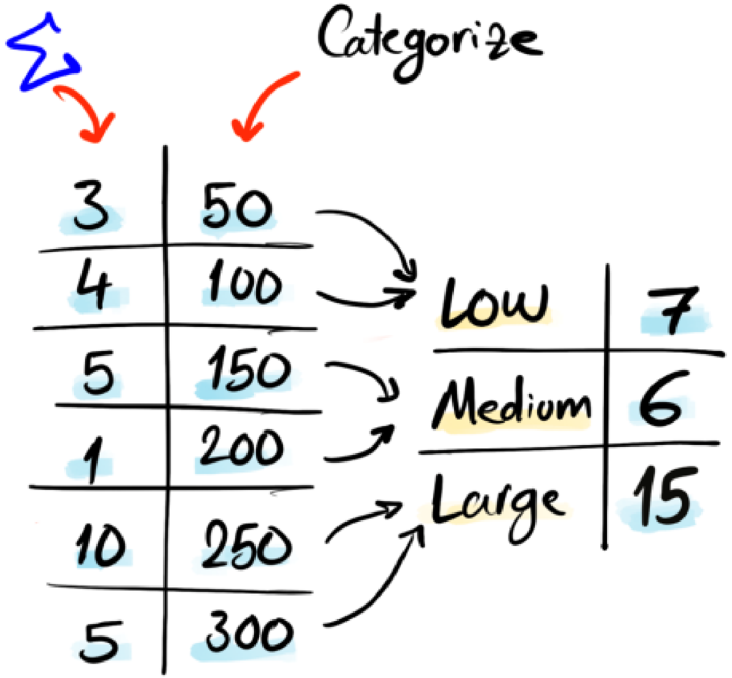
Q1. Which 5 products Generating the Highest Revenue?
Q2. Which 5 products Generating the Highest Revenue? (Using Window & Rank Functions)
Q3. What are the 5 worst-performing products in terms of sales?
Q4. Find the top 10 customers who have generated the highest revenue
Q5. The 3 customers with the fewest orders placed

Purpose:
- To rank items (e.g., products, customers) based on performance or other metrics.
- To identify top performers or laggards.

SQL Functions Used:
- Window Ranking Functions: RANK(), DENSE_RANK(), ROW_NUMBER(), TOP
- Clauses: GROUP BY, ORDER BY



Data Segmentation

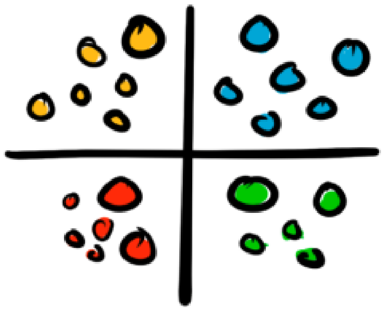


CASE WHEN STATEMENT

[Measure] By [Measure]

Total Products By Sales Range

Total Customers By Age



Activity - Data Segmentation Analysis

Q1. Segment products into cost ranges (0-100, 100-500, 500-1000, & 1000 above) and count how many products fall into each segment.

Q2. Group customers into three segments based on their age & spending behavior and find the total number of customers by each group.
(Lifespan >12 Months & Spending >5000 – “VIP”, Lifespan >12 Months & Spending <5000 – “Regular”, Else –”New”)

Purpose:
- To group data into meaningful categories for targeted insights.
- For customer segmentation, product categorization, or regional analysis.

SQL Functions Used:
- CASE: Defines custom segmentation logic.
- GROUP BY: Groups data into segments.

Final Report Creation

**Report 1 - ** Create Customer Report

Highlights (Key KPIs):

1. Gathers all essential fields such as Customer key, names, ages, and transaction details.
2. Segments customers into categories (VIP, Regular, New) and age groups.
3. Aggregates customer-level metrics:
   - total orders
   - total sales
   - total quantity purchased
   - total products
   - lifespan (in months)
4. Calculates valuable KPIs:
   - recency (months since last order)
   - average order value
   - average monthly spend

Purpose: This report consolidates key customer metrics and behaviors

**Report 2 - ** Create Product Report

Highlights (Key KPIs):

1. Gathers essential fields such as Product Key, product name, category, subcategory, and cost.
2. Segments products by revenue to identify High-Performers, Mid-Range, or Low-Performers.
3. Aggregates product-level metrics:
   - total orders
   - total sales
   - total quantity sold
   - total customers (unique)
   - lifespan (in months)
4. Calculates valuable KPIs:
   - recency (months since last sale)
   - average order revenue (AOR)
   - average monthly revenue

Purpose: This report consolidates key product metrics and behaviors.